

7.1 设备主题宽表

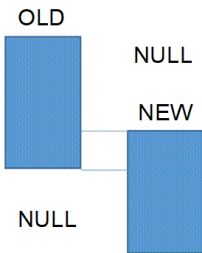
设备主题宽表

1) 建表语句

```
drop table if exists dwt_uv_topic;
create external table dwt_uv_topic
(
  `mid_id` string COMMENT '设备唯一标识',
  `user_id` string COMMENT '用户标识',
  `version_code` string COMMENT '程序版本号',
  `version_name` string COMMENT '程序版本名',
  `lang` string COMMENT '系统语言',
  `source` string COMMENT '渠道号',
  `os` string COMMENT '安卓系统版本',
  `area` string COMMENT '区域',
  `model` string COMMENT '手机型号',
  `brand` string COMMENT '手机品牌',
  `sdk_version` string COMMENT 'sdkVersion',
  `gmail` string COMMENT 'gmail',
  `height_width` string COMMENT '屏幕宽高',
  `app_time` string COMMENT '客户端日志产生时的时间',
  `network` string COMMENT '网络模式',
  `lng` string COMMENT '经度',
  `lat` string COMMENT '纬度',
  `login_date_first` string comment '首次活跃时间',
  `login_date_last` string comment '末次活跃时间',
  `login_day_count` bigint comment '当日活跃次数',
  `login_count` bigint comment '累积活跃天数'
)
stored as parquet
location '/warehouse/gmall/dwt/dwt_uv_topic';
```

2) 数据装载

```
insert overwrite table dwt_uv_topic
select
  nv(new.mid_id,old.mid_id),
  nv(new.user_id,old.user_id),
  nv(new.version_code,old.version_code),
  nv(new.version_name,old.version_name),
  nv(new.lang,old.lang),
  nv(new.source,old.source),
  nv(new.os,old.os),
  nv(new.area,old.area),
  nv(new.model,old.model),
  nv(new.brand,old.brand),
  nv(new.sdk_version,old.sdk_version),
  nv(new.gmail,old.gmail),
  nv(new.height_width,old.height_width),
  nv(new.app_time,old.app_time),
  nv(new.network,old.network),
  nv(new.lng,old.lng),
  nv(new.lat,old.lat),
  if(old.mid_id is null,'2021-08-16',old.login_date_first),
  if(new.mid_id is not null,'2021-08-16',old.login_date_last),
  if(new.mid_id is not null, new.login_count,0),
  nv(old.login_count,0)+if(new.login_count>0,1,0)
from
(
  select * from dwt_uv_topic
)old
full outer join
(
  select * from dws_uv_detail_daycount where dt='2021-08-16'
)new
on old.mid_id=new.mid_id;
```



1) 建表语句

```
hive (gmall)>

drop table if exists dwt_uv_topic;

create external table dwt_uv_topic
(
  `mid_id` string COMMENT '设备唯一标识',

  `model` string COMMENT '手机型号',

  `brand` string COMMENT '手机品牌',

  `login_date_first` string comment '首次活跃时间',

  `login_date_last` string comment '末次活跃时间',

  `login_count` bigint comment '累积活跃天数'
)

stored as parquet

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/dwt/dwt_uv_topic';
```

2) 数据装载

```
hive (gmall)>

insert overwrite table dwt_uv_topic
```

```

select

    nvl(new.mid_id,old.mid_id)-----如果new.mid_id为空就返回old.mid_id，不为空返回
new.mid_id

    nvl(new.model,old.model),-----如果new.model为空就返回old.model，不为空返回new.model

    nvl(new.brand,old.brand),----如果new.brand为空就返回old.brand，不为空返回new.brand

    nvl(old.login_date_first,'2021-08-16'),---如果old.login_date_first为空就返回2021-
08-16，不为空返回old.login_date_first
    if(new.mid_id is not null,'2021-08-16',old.login_date_last),如果new.mid_id 不为
空，这返回2021-08-16，否则返回old.login_date_last

    nvl(old.login_count,0)+if(new.mid_id is not null,1,0)-----old.login_count为空返
回0，加上 new.mid_id不为空返回1，没有就为0

from

(
    select *

    from dwt_uv_topic  -----设备主题宽表，现在表里啥也没有

)old

full outer join

(
    select *

    from dws_uv_detail_daycount -----dws每日设备行为表

    where dt='2021-08-16' -----筛选日期

)new

on old.mid_id=new.mid_id;

```

3) 查询加载结果

```
hive (gmall)> select * from dwt_uv_topic limit 5;
```

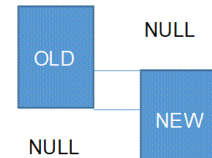
会员主题宽表

1) 建表语句

```
drop table if exists dwt_user_topic;
create external table dwt_user_topic
(
  user_id string comment '用户id',
  login_date_first string comment '首次登录时间',
  login_date_last string comment '末次登录时间',
  login_count bigint comment '累积登录天数',
  login_last_30d_count bigint comment '最近30日登录天数',
  order_date_first string comment '首次下单时间',
  order_date_last string comment '末次下单时间',
  order_count bigint comment '累积下单次数',
  order_amount decimal(16,2) comment '累积下单金额',
  order_last_30d_count bigint comment '最近30日下单次数',
  order_last_30d_amount decimal(16,2) comment '最近30日下单金额',
  payment_date_first string comment '首次支付时间',
  payment_date_last string comment '末次支付时间',
  payment_count decimal(16,2) comment '累积支付次数',
  payment_amount decimal(16,2) comment '累积支付金额',
  payment_last_30d_count decimal(16,2) comment '最近30日支付次数',
  payment_last_30d_amount decimal(16,2) comment '最近30日支付金额'
) COMMENT '用户主题宽表'
stored as parquet
location '/warehouse/gmall/dwt/dwt_user_topic/'
tblproperties ("parquet.compression"="lzo");
```

2) 数据装载

```
insert overwrite table dwt_user_topic
select
  nvl(new.user_id,old.user_id),
  if(old.login_date_first is null and new.login_count>0,'2021-08-16',old.login_date_first),
  if(new.login_count=0,'2021-08-16',old.login_date_last),
  nvl(old.login_count,0)+if(new.login_count>0,1,0),
  nvl(new.login_last_30d_count,0),
  if(old.order_date_first is null and new.order_count>0,'2021-08-16',old.order_date_first),
  if(new.order_count=0,'2021-08-16',old.order_date_last),
  nvl(old.order_count,0)+nvl(new.order_count,0),
  nvl(old.order_amount,0)+nvl(new.order_amount,0),
  nvl(new.order_last_30d_count,0),
  nvl(new.order_last_30d_amount,0),
  if(old.payment_date_first is null and new.payment_count>0,'2021-08-16',old.payment_date_first),
  nvl(old.payment_count,0)+nvl(new.payment_count,0),
  nvl(old.payment_amount,0)+nvl(new.payment_amount,0),
  nvl(new.payment_last_30d_count,0),
  nvl(new.payment_last_30d_amount,0)
from dwt_user_topic old
full outer join
(
  select
    user_id,
    sum(if(dt='2021-08-16',login_count,0)) login_count 登录次数
    sum(if(dt='2021-08-16',order_count,0)) order_count 下单数
    sum(if(dt='2021-08-16',order_amount,0)) order_amount 下单金额
    sum(if(dt='2021-08-16',payment_count,0)) payment_count 支付次数
    sum(if(dt='2021-08-16',payment_amount,0)) payment_amount 支付金额
    sum(if(order_count>0,1,0)) login_last_30d_count 最近30日登录天数
    sum(order_count) order_last_30d_count 最近30日下单次数
    sum(order_amount) order_last_30d_amount 最近30日下单金额
    sum(payment_count) payment_last_30d_count 最近30日支付次数
    sum(payment_amount) payment_last_30d_amount 最近30日支付金额
  from dws_user_action_daycount
  where dt>=date_add('2021-08-16',-30)
  group by user_id
)new on old.user_id=new.user_id;
```



1) 建表语句

```
hive (gmall)>
```

```
drop table if exists dwt_user_topic;
```

```
create external table dwt_user_topic
```

```
(
```

```
  user_id string comment '用户id',
```

```
  login_date_first string comment '首次登录时间',
```

```
  login_date_last string comment '末次登录时间',
```

```
  login_count bigint comment '累积登录天数',
```

```
  login_last_30d_count bigint comment '最近30日登录天数',
```

```
  order_date_first string comment '首次下单时间',
```

```
  order_date_last string comment '末次下单时间',
```

```
  order_count bigint comment '累积下单次数',
```

```
  order_amount decimal(16,2) comment '累积下单金额',
```

```
  order_last_30d_count bigint comment '最近30日下单次数',
```

```
  order_last_30d_amount decimal(16,2) comment '最近30日下单金额',
```

```
  payment_date_first string comment '首次支付时间',
```

```
  payment_date_last string comment '末次支付时间',
```

```

payment_count decimal(16,2) comment '累积支付次数',

payment_amount decimal(16,2) comment '累积支付金额',

payment_last_30d_count decimal(16,2) comment '最近30日支付次数',

payment_last_30d_amount decimal(16,2) comment '最近30日支付金额'

)COMMENT '用户主题宽表'

stored as parquet

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/dwt/dwt_user_topic/'

tblproperties ("parquet.compression"="lzo");

```

2) 数据装载

```

insert overwrite table dwt_user_topic
select
    -- 用户ID处理，新表id为空返回老表，不为空返回新表id
    nvl(new.user_id, old.user_id) as user_id,

    -- 首次登录时间（新用户取当天，老用户保留原值）
    if(old.login_date_first is null and new.user_id is not null, '2021-08-16',
old.login_date_first),

    -- 最近登录时间（当天有行为则更新为当天）
    if(new.user_id is not null, '2021-08-16', old.login_date_last),

    -- 累计登录天数（历史累计 + 当天是否登录）
    nvl(old.login_count, 0) + if(new.user_id is not null, 1, 0),

    -- 最近30天登录天数（直接取自当天计算的30日聚合值）
    nvl(new.login_last_30d_count, 0),

    -- 首次下单时间（新增下单用户取当天）
    if(old.order_date_first is null and new.order_count > 0, '2021-08-16',
old.order_date_first),

    -- 最近下单时间（当天有下单则更新）
    if(new.order_count > 0, '2021-08-16', old.order_date_last),

    -- 累计下单次数（历史 + 当天）
    nvl(old.order_count, 0) + nvl(new.order_count, 0),

    -- 累计下单金额（历史 + 当天）
    nvl(old.order_amount, 0) + nvl(new.order_amount, 0),

    -- 最近30天下单次数（直接取自30日聚合）
    nvl(new.order_last_30d_count, 0),

    -- 最近30天下单金额（直接取自30日聚合）
    nvl(new.order_last_30d_amount, 0),

```

```

-- 首次支付时间（新增支付用户取当天）
if(old.payment_date_first is null and new.payment_count > 0, '2021-08-16',
old.payment_date_first),

-- 最近支付时间（当天有支付则更新）
if(new.payment_count > 0, '2021-08-16', old.payment_date_last),

-- 累计支付次数（历史 + 当天）
nvl(old.payment_count, 0) + nvl(new.payment_count, 0),

-- 累计支付金额（历史 + 当天）
nvl(old.payment_amount, 0) + nvl(new.payment_amount, 0),

-- 最近30天支付次数（直接取自30日聚合）
nvl(new.payment_last_30d_count, 0),

-- 最近30天支付金额（直接取自30日聚合）
nvl(new.payment_last_30d_amount, 0)

from
-- 全外连接：合并历史数据(old)与当天增量(new)
dwt_user_topic old
full outer join
(
    select
        user_id,
        -- 当天行为指标
        sum(if(dt='2021-08-16', order_count, 0)) as order_count,
        sum(if(dt='2021-08-16', order_amount, 0)) as order_amount,
        sum(if(dt='2021-08-16', payment_count, 0)) as payment_count,
        sum(if(dt='2021-08-16', payment_amount, 0)) as payment_amount,
        -- 最近30天登录天数（统计周期内登录天数）
        sum(if(login_count > 0, 1, 0)) as login_last_30d_count,
        -- 最近30天下单指标
        sum(order_count) as order_last_30d_count,
        sum(order_amount) as order_last_30d_amount,
        sum(payment_count) as payment_last_30d_count,
        sum(payment_amount) as payment_last_30d_amount
    from dws_user_action_daycount -- 来源：DWS层日粒度用户行为表
    where dt >= date_add('2021-08-16', -30) -- 取30天内的数据
    group by user_id
) new
on old.user_id = new.user_id;

```

7.3 商品主题宽表

1) 建表语句

```

hive (gmall)>

drop table if exists dwt_sku_topic;

create external table dwt_sku_topic

(

```

```
sku_id string comment 'sku_id',

spu_id string comment 'spu_id',

order_last_30d_count bigint comment '最近30日被下单次数',

order_last_30d_num bigint comment '最近30日被下单件数',

order_last_30d_amount decimal(16,2) comment '最近30日被下单金额',

order_count bigint comment '累积被下单次数',

order_num bigint comment '累积被下单件数',

order_amount decimal(16,2) comment '累积被下单金额',

payment_last_30d_count bigint comment '最近30日被支付次数',

payment_last_30d_num bigint comment '最近30日被支付件数',

payment_last_30d_amount decimal(16,2) comment '最近30日被支付金额',

payment_count bigint comment '累积被支付次数',

payment_num bigint comment '累积被支付件数',

payment_amount decimal(16,2) comment '累积被支付金额',

refund_last_30d_count bigint comment '最近三十日退款次数',

refund_last_30d_num bigint comment '最近三十日退款件数',

refund_last_30d_amount decimal(10,2) comment '最近三十日退款金额',

refund_count bigint comment '累积退款次数',

refund_num bigint comment '累积退款件数',

refund_amount decimal(10,2) comment '累积退款金额',

cart_last_30d_count bigint comment '最近30日被加入购物车次数',

cart_count bigint comment '累积被加入购物车次数',

favor_last_30d_count bigint comment '最近30日被收藏次数',

favor_count bigint comment '累积被收藏次数',

appraise_last_30d_good_count bigint comment '最近30日好评数',

appraise_last_30d_mid_count bigint comment '最近30日中评数',

appraise_last_30d_bad_count bigint comment '最近30日差评数',

appraise_last_30d_default_count bigint comment '最近30日默认评价数',
```

```

appraise_good_count bigint comment '累积好评数',

appraise_mid_count bigint comment '累积中评数',

appraise_bad_count bigint comment '累积差评数',

appraise_default_count bigint comment '累积默认评价数'

)COMMENT '商品主题宽表'

stored as parquet

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/dwt/dwt_sku_topic/'

tblproperties ("parquet.compression"="lzo");

-- 全量覆盖写入（每日调度刷新）
insert overwrite table dwt_sku_topic
select
    -- SKU ID处理（优先取当日增量数据）
    nv1(new.sku_id, old.sku_id) as sku_id,
    -- 关联商品SPU信息
    sku_info.spu_id,
    -- 近30日订单指标（直接取自当日计算的滚动聚合）
    nv1(new.order_count30, 0),
    nv1(new.order_num30, 0),
    nv1(new.order_amount30, 0),
    -- 历史累计订单（旧数据 + 当日增量）
    nv1(old.order_count, 0) + nv1(new.order_count, 0),
    nv1(old.order_num, 0) + nv1(new.order_num, 0),
    nv1(old.order_amount, 0) + nv1(new.order_amount, 0),
    -- 支付指标（处理逻辑同上）
    nv1(new.payment_count30, 0),
    nv1(new.payment_num30, 0),
    nv1(new.payment_amount30, 0),
    nv1(old.payment_count, 0) + nv1(new.payment_count, 0),
    nv1(old.payment_num, 0) + nv1(new.payment_num, 0), -- 注意此处应为
new.payment_num
    nv1(old.payment_amount, 0) + nv1(new.payment_amount, 0),
    -- 退款指标
    nv1(new.refund_count30, 0),
    nv1(new.refund_num30, 0),
    nv1(new.refund_amount30, 0),
    nv1(old.refund_count, 0) + nv1(new.refund_count, 0),
    nv1(old.refund_num, 0) + nv1(new.refund_num, 0),
    nv1(old.refund_amount, 0) + nv1(new.refund_amount, 0),
    -- 用户行为指标
    nv1(new.cart_count30, 0),
    nv1(old.cart_count, 0) + nv1(new.cart_count, 0),
    nv1(new.favor_count30, 0),
    nv1(old.favor_count, 0) + nv1(new.favor_count, 0),
    -- 近30日评价
    nv1(new.appraise_good_count30, 0),
    nv1(new.appraise_mid_count30, 0),
    nv1(new.appraise_bad_count30, 0),

```

```

    nvl(new.appraise_default_count30, 0),
    -- 历史累计评价
    nvl(old.appraise_good_count, 0) + nvl(new.appraise_good_count, 0),
    nvl(old.appraise_mid_count, 0) + nvl(new.appraise_mid_count, 0),
    nvl(old.appraise_bad_count, 0) + nvl(new.appraise_bad_count, 0),
    nvl(old.appraise_default_count, 0) + nvl(new.appraise_default_count, 0)
from
    -- 旧数据（全量历史数据）
    (select * from dwt_sku_topic) old
full outer join
    -- 新数据（当日增量 + 30日滚动聚合）
    (
        select
            sku_id,
            -- 当日行为指标
            sum(if(dt='2021-08-16', order_count, 0)) as order_count,
            sum(if(dt='2021-08-16', order_num, 0)) as order_num,
            sum(if(dt='2021-08-16', order_amount, 0)) as order_amount,
            sum(if(dt='2021-08-16', payment_count, 0)) as payment_count,
            sum(if(dt='2021-08-16', payment_num, 0)) as payment_num,
            sum(if(dt='2021-08-16', payment_amount, 0)) as payment_amount,
            sum(if(dt='2021-08-16', refund_count, 0)) as refund_count,
            sum(if(dt='2021-08-16', refund_num, 0)) as refund_num,
            sum(if(dt='2021-08-16', refund_amount, 0)) as refund_amount,
            sum(if(dt='2021-08-16', cart_count, 0)) as cart_count,
            sum(if(dt='2021-08-16', favor_count, 0)) as favor_count,
            sum(if(dt='2021-08-16', appraise_good_count, 0)) as
appraise_good_count,
            sum(if(dt='2021-08-16', appraise_mid_count, 0)) as
appraise_mid_count,
            sum(if(dt='2021-08-16', appraise_bad_count, 0)) as
appraise_bad_count,
            sum(if(dt='2021-08-16', appraise_default_count, 0)) as
appraise_default_count,
            -- 近30日滚动聚合指标
            sum(order_count) as order_count30,
            sum(order_num) as order_num30,
            sum(order_amount) as order_amount30,
            sum(payment_count) as payment_count30,
            sum(payment_num) as payment_num30,
            sum(payment_amount) as payment_amount30,
            sum(refund_count) as refund_count30,
            sum(refund_num) as refund_num30,
            sum(refund_amount) as refund_amount30,
            sum(cart_count) as cart_count30,
            sum(favor_count) as favor_count30,
            sum(appraise_good_count) as appraise_good_count30,
            sum(appraise_mid_count) as appraise_mid_count30,
            sum(appraise_bad_count) as appraise_bad_count30,
            sum(appraise_default_count) as appraise_default_count30
        from dws_sku_action_daycount -- 来源: DWS层日粒度商品行为表
        where dt >= date_add('2021-08-16', -30) -- 取30天内的数据
        group by sku_id
    ) new
on old.sku_id = new.sku_id
-- 关联商品SPU维度表（获取SPU信息）

```

```

left join
    (select * from dwd_dim_sku_info where dt='2021-08-16') sku_info  -- 维度表取当日最新快照
on nvl(new.sku_id, old.sku_id) = sku_info.id;  -- 处理新旧数据中的sku_id

```

8.1.1 活跃设备数（日、周、月）

需求定义：

日活：当日活跃的设备数

周活：当周活跃的设备数

月活：当月活跃的设备数

```

hive (gmall)> -----建表
drop table if exists ads_uv_count;
create external table ads_uv_count(
    `dt` string COMMENT '统计日期',
    `day_count` bigint COMMENT '当日用户数量',
    `wk_count` bigint COMMENT '当周用户数量',
    `mn_count` bigint COMMENT '当月用户数量',
    `is_weekend` string COMMENT 'Y,N是否是周末,用于得到本周最终结果',
    `is_monthend` string COMMENT 'Y,N是否是月末,用于得到本月最终结果'
) COMMENT '活跃设备数'
row format delimited fields terminated by '\t'
location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_uv_count/';

hive (gmall)>
insert into table ads_uv_count
select

    '2021-08-16' dt,--初始日期
    daycount.ct,--日活设备数
    wkcount.ct,--周活设备数
    mncount.ct,--月活设备数
    if(date_add(next_day('2021-08-16','MO'),-1)='2021-08-16','Y','N') ,
    if(last_day('2021-08-16')='2021-08-16','Y','N')
from
(
    select
    '2021-08-16' dt,
    count(*) ct-----计算2021-08-16的登录人数
from dwt_uv_topic -----返回dwt层设备主题表
where login_date_last='2021-08-16' -----筛选登出时间为2021-08-16

)daycount join -----连接
(
    select
    '2021-08-16' dt,
    count (*) ct -----计算周活的设备数
    from dwt_uv_topic -----返回dwt层设备主题表
    where login_date_last>=date_add(next_day('2021-08-16','MO'),-7) -----返回周一再
    减去7天, 上周一为开始

```

```

        and login_date_last<= date_add(next_day('2021-08-16','MO'),-1) -----返回这周
        天, 为时间结束
    ) wkcount on daycount.dt=wkcount.dt
join
(
    select
    '2021-08-16' dt,
    count (*) ct -----计算周活的设备数
    from dwt_uv_topic -----返回dwt层设备主题表
where date_format(login_date_last,'yyyy-MM')=date_format('2021-08-16','yyyy-MM')-
--- 把月份定在本月 这里是8月
)mncount on daycount.dt=mncount.dt;

```

8.1.2 每日新增设备

1) 建表语句

```

hive (gmall)>

drop table if exists ads_new_mid_count;

create external table ads_new_mid_count

(

    `create_date`    string comment '创建时间' ,

    `new_mid_count`  BIGINT comment '新增设备数量'

) COMMENT '每日新增设备信息数量'

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_new_mid_count/';

```

2) 导入数据

```

hive (gmall)>

insert into table ads_new_mid_count

select

    login_date_first,-----第一次登录时间

    count(*)-----计算新增

from dwt_uv_topic-----返回dwt层设备主题表 获取数据

where login_date_first='2021-08-16'

group by login_date_first;

```

3) 查询导入数据

```
hive (gmall)> select * from ads_new_mid_count;
```

8.1.3 沉默用户数

需求定义：

沉默用户：只在安装当天启动过，且启动时间是在7天前

1) 建表语句

```
hive (gmall)>
```

```
drop table if exists ads_silent_count;
```

```
create external table ads_silent_count(
```

```
    `dt` string COMMENT '统计日期',
```

```
    `silent_count` bigint COMMENT '沉默设备数'
```

```
)
```

```
row format delimited fields terminated by '\t'
```

```
location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_silent_count';
```

2) 导入2021-08-16数据

```
hive (gmall)>
```

```
insert into table ads_silent_count
```

```
select
```

```
    '2021-08-16',
```

```
    count(*) -----计算沉默用户人数
```

```
from dwt_uv_topic -----返回dwt层的设备主题宽表
```

```
where login_date_first=login_date_last-----筛选第一次登录日期等于最后登录日期
```

```
and login_date_last<=date_add('2021-08-16',-7);----并且最后登录日期小于等于上周，人话就是第一次登陆后连续七天没登陆过
```

3) 查询导入数据

```
hive (gmall)> select * from ads_silent_count;
```

8.1.4 本周回流用户数

需求定义：

本周回流用户：上周末活跃，本周活跃的设备，且不是本周新增设备

1) 建表语句

```
hive (gmall)>

drop table if exists ads_back_count;

create external table ads_back_count(

    `dt` string COMMENT '统计日期',

    `wk_dt` string COMMENT '统计日期所在周',

    `wastage_count` bigint COMMENT '回流设备数'

)

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_back_count';
```

2) 导入数据:

```
hive (gmall)>

insert into table ads_back_count

select

'2021-08-16',

concat(date_add(next_day('2021-08-16','MO'),-7),'_', date_add(next_day('2021-08-16','MO'),-1)),  -----将这周一的日期和这周末的日期拼接起来

count(*)

from

(

select

    mid_id

from dwt_uv_topic  -----返回dwt层的设备主题宽表

where login_date_last>=date_add(next_day('2021-08-16','MO'),-7)
-----最后登录日期大于等于这周一并且小于等于这周末；第一次登录日期小于这周一
and login_date_last<= date_add(next_day('2021-08-16','MO'),-1)

and login_date_first<date_add(next_day('2021-08-16','MO'),-7)
```

```

-----人话就是在这周登录过并且不是新用户
)current_wk ----这周

left join -----左外连接

(

select

mid_id

from dws_uv_detail_daycount -----返回dws层的每日设备行为表

where dt>=date_add(next_day('2021-08-16','MO'),-7*2) ----筛选上周的时间

and dt<= date_add(next_day('2021-08-16','MO'),-7-1)

group by mid_id

)last_wk

on current_wk.mid_id=last_wk.mid_id 然后有用户id连接

where last_wk.mid_id is null; 筛选上周没有登录过的用户，然后这周登录过的为回流用户，新用户不算

3) 查询结果

hive (gmall)> select * from ads_back_count;

```

8.1.5 流失用户数

需求定义：

流失用户：最近7天未活跃的设备

```

1) 建表语句

hive (gmall)>

drop table if exists ads_wastage_count;

create external table ads_wastage_count(

    `dt` string COMMENT '统计日期',

    `wastage_count` bigint COMMENT '流失设备数'

)

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_wastage_count';

2) 导入2021-08-16数据

```

```
hive (gmall)>

insert into table ads_wastage_count

select

    '2021-08-16',

    count(*)

from

(

    select

        mid_id

    from dwt_uv_topic-----返回dwt设备主题宽表

    where login_date_last<=date_add('2021-08-16',-7)-----最后登录时间不为本周的用户

    group by mid_id

)t1;
```

3) 查询结果

```
hive (gmall)> select * from ads_wastage_count;
```

8.1.6 留存率

用户留存

留存用户：某段时间内的新增用户（活跃用户），经过一段时间后，又继续使用应用的被认作是留存用户；

留存率：留存用户占当时新增用户（活跃用户）的比例即是留存率。

例如，2月10日新增用户100，这100人在2月11日启动过应用的有30人，2月12日启动过应用的有25人，2月13日启动过应用的有32人；

则2月10日新增用户次日的留存率是 $30/100 = 30\%$ ，两日留存率是 $25/100=25\%$ ，三日留存率是 $32/100=32\%$ 。

时间	新增用户	1天后	2天后	3天后
2021-02-10	100	30% (2-11)	25% (2-12)	32% (2-13)
2021-02-11	200	20% (2-12)	15% (2-13)	
2021-02-12	100	25% (2-13)		
2021-02-13				

1) 建表语句

```
hive (gmall)>
```

```

drop table if exists ads_user_retention_day_rate;

create external table ads_user_retention_day_rate
(
    `stat_date`      string comment '统计日期',
    `create_date`    string comment '设备新增日期',
    `retention_day`   int comment '截止当前日期留存天数',
    `retention_count` bigint comment '留存数量',
    `new_mid_count`   bigint comment '设备新增数量',
    `retention_ratio` decimal(10,2) comment '留存率'
) COMMENT '每日用户留存情况'

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-
3.1.2/warehouse/gmall.db/ads/ads_user_retention_day_rate/';

```

2) 导入数据

```

hive (gmall)>

insert into table ads_user_retention_day_rate

select

    '2021-08-16',--统计日期

    date_add('2021-08-16',-1),--新增日期

    1,--留存天数

    sum(if(login_date_first=date_add('2021-08-16',-1) and login_date_last='2021-08-16',1,0)),--2021-08-15的1日留存数

    sum(if(login_date_first=date_add('2021-08-16',-1),1,0)),--2021-08-15新增

    sum(if(login_date_first=date_add('2021-08-16',-1) and login_date_last='2021-08-16',1,0))/sum(if(login_date_first=date_add('2021-08-16',-1),1,0))*100----当日的留存率

from dwt_uv_topic

union all

```

```

select

    '2021-08-16',--统计日期

    date_add('2021-08-16',-2),--新增日期

    2,--留存天数

    sum(if(login_date_first=date_add('2021-08-16',-2) and login_date_last='2021-08-16',1,0)),--2021-08-14的2日留存数

    sum(if(login_date_first=date_add('2021-08-16',-2),1,0)),--2021-08-14新增

    sum(if(login_date_first=date_add('2021-08-16',-2) and login_date_last='2021-08-16',1,0))/sum(if(login_date_first=date_add('2021-08-16',-2),1,0))*100----两天的留存率

from dwt_uv_topic

```

union all

```

select

    '2021-08-16',--统计日期

    date_add('2021-08-16',-3),--新增日期

    3,--留存天数

    sum(if(login_date_first=date_add('2021-08-16',-3) and login_date_last='2021-08-16',1,0)),--2021-08-13的3日留存数

    sum(if(login_date_first=date_add('2021-08-16',-3),1,0)),--2021-08-13新增

    sum(if(login_date_first=date_add('2021-08-16',-3) and login_date_last='2021-08-16',1,0))/sum(if(login_date_first=date_add('2021-08-16',-3),1,0))*100---三天的留存率

from dwt_uv_topic;

```

3) 查询导入数据

```
hive (gmall)>select * from ads_user_retention_day_rate;
```

8.2 会员主题

8.2.1 会员主题信息

1) 建表

```
hive (gmall)>

drop table if exists ads_user_topic;

create external table ads_user_topic(

    `dt` string COMMENT '统计日期',

    `day_users` string COMMENT '活跃会员数',

    `day_new_users` string COMMENT '新增会员数',

    `day_new_payment_users` string COMMENT '新增消费会员数',

    `payment_users` string COMMENT '总付费会员数',

    `users` string COMMENT '总会员数',

    `day_users2users` decimal(10,2) COMMENT '会员活跃率',

    `payment_users2users` decimal(10,2) COMMENT '会员付费率',

    `day_new_users2users` decimal(10,2) COMMENT '会员新鲜度'

) COMMENT '会员主题信息表'

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_user_topic';
```

2) 导入数据

```
hive (gmall)>

-- 全量插入当日统计结果
insert into table ads_user_topic
select
    '2021-08-16' as dt, -- 固定统计日期（应参数化）
    sum(if(login_date_last='2021-08-16', 1, 0)) as day_users, -- 当日活跃：末次登录
    为当天
    sum(if(login_date_first='2021-08-16', 1, 0)) as day_new_users, -- 当日新增：首
    次登录为当天
    sum(if(payment_date_first='2021-08-16', 1, 0)) as day_new_payment_users, --
    当日首付
    sum(if(payment_count > 0, 1, 0)) as payment_users, -- 历史累计付费会员
    count(*) as users, -- 当前总会员数
    -- 计算比率指标（保留两位小数）
    round(sum(if(login_date_last='2021-08-16',1,0)) / count(*), 2) as
    day_users2users, -- 活跃率
```

```

round(sum(if(payment_count>0,1,0)) / count(*), 2) as payment_users2users, -- 付费率
round(sum(if(login_date_first='2021-08-16',1,0)) /
sum(if(login_date_last='2021-08-16',1,0)), 2) as day_new_users2users -- 新增占比日活
from dwt_user_topic; -- 来源: DWT层用户主题宽表

3) 查询数据

hive (gmall)> select * from ads_user_topic;

```

8.2.2 漏斗分析

统计“首页->购物车->下单->支付”的转化率

思路：统计各个行为的人数，然后计算比值。

1) 建表语句

```

hive (gmall)>

drop table if exists ads_user_action_convert_day;

create external table ads_user_action_convert_day(

    `dt` string COMMENT '统计日期',

    `total_visitor_m_count` bigint COMMENT '总访问人数',

    `cart_u_count` bigint COMMENT '加入购物车的人数',

    `visitor2cart_convert_ratio` decimal(10,2) COMMENT '访问到加入购物车转化率',

    `order_u_count` bigint COMMENT '下单人数',

    `cart2order_convert_ratio` decimal(10,2) COMMENT '加入购物车到下单转化率',

    `payment_u_count` bigint COMMENT '支付人数',

    `order2payment_convert_ratio` decimal(10,2) COMMENT '下单到支付的转化率'

) COMMENT '用户行为漏斗分析'

row format delimited fields terminated by '\t'

location '/usr/hive/apache-hive-3.1.2/warehouse/gmall.db/ads/ads_user_action_convert_day/';

```

2) 数据装载

```

hive (gmall)>

-- 插入当日转化率数据
insert into table ads_user_action_convert_day

```

```

select
    '2021-08-16' as dt, -- 统计日期（应参数化）
    uv.day_count as total_visitor_m_count, -- 当日UV（来自访问统计表）
    ua.cart_count as cart_u_count, -- 当日加购人数
    -- 计算各阶段转化率
    round(ua.cart_count / uv.day_count, 2) as visitor2cart_convert_ratio, -- 访问->加购
    ua.order_count as order_u_count, -- 当日下单人数
    round(ua.order_count / ua.cart_count, 2) as cart2order_convert_ratio, -- 加购->下单
    ua.payment_count as payment_u_count, -- 当日支付人数
    round(ua.payment_count / ua.order_count, 2) as order2payment_convert_ratio -- 下单->支付
from
    (
        -- 获取当日行为人数（加购、下单、支付）
        select
            dt,
            sum(if(cart_count > 0, 1, 0)) as cart_count, -- 当日有加购行为的人数
            sum(if(order_count > 0, 1, 0)) as order_count, -- 当日有下单行为的人数
            sum(if(payment_count > 0, 1, 0)) as payment_count -- 当日有支付行为的人数
        from dws_user_action_daycount -- 来源：DWS层用户行为日聚合表
        where dt = '2021-08-16' -- 当日数据
        group by dt
    ) ua
join ads_uv_count uv -- 关联当日UV统计表
on uv.dt = ua.dt; -- 按日期连接

```